

Written Justification

Building Age Classification from Satellite Imagery

GROUP 6

Challenge & Approach. We classify building age (4 classes) from Landsat spectral data, training on Madrid and testing transfer to Amsterdam, cities differing in climate, materials, and architecture. Age classes were derived from *weighted_mean_year* using city-specific boundaries (1960 Madrid, 1945 Amsterdam) plus shared 1984/2004 cutoffs. Building coverage per pixel ranged 15–100% (median ~46%), with every pixel carrying a usable label.

Model Design. Beyond the 6 raw spectral bands, we engineered 5 normalised indices (NDVI, NDBI, UI, MNDWI, BSI), collapsed into 60 temporal features per pixel (overall, early-era, late-era, and year-on-year statistics). Boxplots confirmed the expected pattern and it was that older neighbourhoods show lower vegetation and higher built-up density than modern developments, evidence the engineered features carry real class-separating signal. We chose Random Forest, evaluated via 5-fold×5-repeat cross-validation on Madrid, reaching macro-F1 0.628 ± 0.004 . Adding *class_weight="balanced"* was the single most impactful tuning choice, meaningfully improving recall on the smallest class (Class 4) without materially harming the others. We separately tested trimming to the 20 most important features: Madrid-only cross-validation performance dropped when features were removed, but the reduced model achieved a marginally higher zero-shot Amsterdam score (0.451 vs 0.443) than the full 60-feature model, suggesting some low-importance features add Madrid-specific noise that mildly hinders transfer, even though they help fit Madrid itself. This is a useful, if modest, finding: optimising for the training city is not the same objective as optimising for transfer.

Transfer Strategy. The Madrid-trained model is evaluated on Amsterdam in two stages. First, zero-shot: predicting directly with zero Amsterdam labels, isolating the raw domain gap between cities. Second, few-shot adaptation: using small, labelled Amsterdam samples (5, 25, 50, 100, 200 per class), we compute a class "prototype" (average feature vector) from the support set and classify remaining pixels by nearest prototype. This directly targets the challenge's central question - whether spectral representations learned in one city generalise to another with minimal local relabelling, rather than requiring full retraining.

F1 Interpretation. Zero-shot Amsterdam macro-F1 was 0.443, substantially below the Madrid score of 0.628, quantifying a real, non-trivial domain gap consistent with the two cities' differing architecture and climate. This gap closed steadily with more labelled data: 0.544 at 5 samples/class, rising to 0.596 at 25, 0.605 at 50, 0.611 at 100, and 0.615 at 200/class. The steepest gain occurs between 5 and 25 samples, after that, returns diminish noticeably, suggesting the prototype method extracts most of its useful signal from a small support set rather than requiring large-scale Amsterdam labelling. This is an encouraging result for low-data transfer: even 25 labelled examples per class recover roughly 70% of the total improvement achieved at 200/class. Taken together, these results indicate that spectral indices engineered for physical meaning, rather than raw reflectance alone provide a representation that meaningfully, if imperfectly, transfers across cities with limited local supervision.

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